Radar Cross Section in DiscoGMPI: Theory, Conventions, and Numerical Evaluation

DiscoGMPI Documentation

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Abstract

This document gives a self-contained description of radar cross section (RCS), the time-harmonic scattering model used for a perfectly electrically conducting sphere, and the numerical procedure implemented in DiscoGMPI. It fixes the physical assumptions, phase convention, angular convention, normalization, near-to-far transformation, time-windowed Fourier extraction, DG surface quadrature, MPI reduction, output files, and validation strategy. The focus is the metallic-sphere workflow implemented in `examples/distributed_metallic_sphere_scattering.jl`, with validation and Mie-series comparison through `examples/validate_coarse_sphere_rcs.jl` and `examples/postprocess_metallic_sphere_mie_rcs.py`.

Contents

1 Purpose of RCS

Radar cross section is a far-field measure of how strongly a target scatters an incident electromagnetic wave into a selected observation direction. It is defined so that a distant receiver can interpret the scattered field as if it had been produced by an equivalent isotropic scatterer with area σ. Large RCS means that the object returns a strong scattered field in that direction; small RCS means that it scatters weakly.

For a monochromatic incident plane wave with electric-field amplitude $|\boldsymbol{E}_{\mathrm{inc}}|$, the far scattered field has the asymptotic form

$$\widehat{\boldsymbol{E}_{\mathrm{scat}}}(\boldsymbol{x}) = \frac{\mathrm{e}^{ikr}}{r}\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) + \mathcal{O}(r^{-2}), \qquad r = |\boldsymbol{x}|, \qquad \widehat{\boldsymbol{r}} = \frac{\boldsymbol{x}}{|\boldsymbol{x}|}. \tag{1}$$

The RCS in direction $\widehat{\boldsymbol{r}}$ is

$$\sigma(\widehat{\boldsymbol{r}}) = \lim_{r\to\infty} 4\pi r^2 \frac{|\widehat{\boldsymbol{E}_{\mathrm{scat}}}(r\widehat{\boldsymbol{r}})|^2}{|\widehat{\boldsymbol{E}_{\mathrm{inc}}}|^2} = 4\pi \frac{|\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}})|^2}{|\widehat{\boldsymbol{E}_{\mathrm{inc}}}|^2}. \tag{2}$$

In DiscoGMPI, the incident amplitude is denoted by A, so $|\widehat{\boldsymbol{E}_{\mathrm{inc}}}| = A$. For a sphere of radius a, postprocessing usually plots the dimensionless normalized RCS

$$\frac{\sigma}{\pi a^2} \tag{3}$$

or its decibel value

$$10\log_{10}\left(\frac{\sigma}{\pi a^2}\right). \tag{4}$$

2 Physical Assumptions

The current DiscoGMPI metallic-sphere RCS workflow assumes:

1. The exterior medium is homogeneous, isotropic, and lossless outside the artificial absorbing layer.

2. The material parameters are scalar constants $\epsilon > 0$ and $\mu > 0$, with wave speed

   $$c = \frac{1}{\sqrt{\epsilon\mu}}, \qquad Z = \sqrt{\frac{\mu}{\epsilon}},$$

   where Z is the wave impedance.

3. The scatterer is a perfectly electrically conducting (PEC) sphere centered in a Cartesian computational box.
4. The numerical unknown is the scattered field. The physical field is reconstructed as incident plus scattered field.
5. The outer computational box is truncated with the nonlinear Cartesian PML available in DiscoGMPI.
6. The RCS diagnostic is extracted from the PEC surface current. The current implementation does not yet use a Huygens box or volume near-to-far transformation.
7. The incident wave is monochromatic and the reported RCS is the Fourier component at the incident frequency. Time-domain transients must be excluded by choosing an appropriate `-rcs-start-time`.

3 Time-Harmonic Convention

DiscoGMPI uses the real incident wave

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{x},t) = A\,\widehat{\boldsymbol{x}}\cos(kz-\omega t) \tag{5}$$

for the default sphere experiment. This is equivalent to the phasor convention

$$\boldsymbol{E}(\boldsymbol{x},t) = \mathrm{Re}\left\{\widehat{\boldsymbol{E}}(\boldsymbol{x})\,\mathrm{e}^{-\mathrm{i}\omega t}\right\}. \tag{6}$$

With this convention, the default incident phasors are

$$\widehat{\boldsymbol{E}_{\mathrm{inc}}}(\boldsymbol{x}) = A\,\widehat{\boldsymbol{x}}\,\mathrm{e}^{\mathrm{i}kz}, \qquad \widehat{\boldsymbol{H}_{\mathrm{inc}}}(\boldsymbol{x}) = \frac{A}{Z}\,\widehat{\boldsymbol{y}}\,\mathrm{e}^{\mathrm{i}kz}. \tag{7}$$

The corresponding real magnetic field is

$$\boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x},t) = \frac{A}{Z}\,\widehat{\boldsymbol{y}}\cos(kz-\omega t). \tag{8}$$

The general plane-wave form is

$$\widehat{\boldsymbol{E}_{\mathrm{inc}}}(\boldsymbol{x}) = A\,\boldsymbol{p}\,\mathrm{e}^{\mathrm{i}k\boldsymbol{d}\cdot\boldsymbol{x}}, \qquad \widehat{\boldsymbol{H}_{\mathrm{inc}}}(\boldsymbol{x}) = \frac{A}{Z}\,(\boldsymbol{d}\times\boldsymbol{p})\,\mathrm{e}^{\mathrm{i}k\boldsymbol{d}\cdot\boldsymbol{x}}, \tag{9}$$

where $\boldsymbol{d}$ is the unit propagation direction, $\boldsymbol{p}$ is the unit electric polarization, and $\boldsymbol{d}\cdot\boldsymbol{p}=0$. The default code values are

$$\boldsymbol{d} = \widehat{\boldsymbol{z}}, \qquad \boldsymbol{p} = \widehat{\boldsymbol{x}}.$$

4 Maxwell Scattering Problem

In the physical exterior domain Ω^+, the total fields satisfy

$$\epsilon\frac{\partial \boldsymbol{E}_{\mathrm{tot}}}{\partial t} = \nabla\times\boldsymbol{H}_{\mathrm{tot}}, \tag{10}$$

$$\mu\frac{\partial \boldsymbol{H}_{\mathrm{tot}}}{\partial t} = -\nabla\times\boldsymbol{E}_{\mathrm{tot}}. \tag{11}$$

In phasor form, using $\mathrm{e}^{-\mathrm{i}\omega t}$,

$$-\mathrm{i}\omega\epsilon\,\widehat{\boldsymbol{E}_{\mathrm{tot}}} = \nabla\times\widehat{\boldsymbol{H}_{\mathrm{tot}}}, \tag{12}$$

$$-\mathrm{i}\omega\mu\,\widehat{\boldsymbol{H}_{\mathrm{tot}}} = -\nabla\times\widehat{\boldsymbol{E}_{\mathrm{tot}}}. \tag{13}$$

The wavenumber and angular frequency are related by

$$k = \frac{\omega}{c} = \omega\sqrt{\epsilon\mu} = \frac{2\pi}{\lambda}. \tag{14}$$

The total field is decomposed as

$$\boldsymbol{E}_{\text{tot}} = \boldsymbol{E}_{\text{inc}} + \boldsymbol{E}_{\text{scat}}, \qquad \boldsymbol{H}_{\text{tot}} = \boldsymbol{H}_{\text{inc}} + \boldsymbol{H}_{\text{scat}}. \tag{15}$$

The numerical field stored by the driver is

$$\boldsymbol{U}_h = \left(\boldsymbol{E}_{\text{scat},h}, \boldsymbol{H}_{\text{scat},h}\right). \tag{16}$$

This is useful because the incident field is known analytically everywhere, including on the metallic surface and in the PML region.

4.1 PEC Boundary Condition

On the metallic sphere Γ_{PEC}, the total tangential electric field must vanish:

$$\boldsymbol{n} \times \boldsymbol{E}_{\text{tot}} = \boldsymbol{0} \qquad \text{on } \Gamma_{\text{PEC}}. \tag{17}$$

Using the scattered-field decomposition, the condition imposed by the code is

$$\boldsymbol{n} \times \boldsymbol{E}_{\text{scat}} = -\boldsymbol{n} \times \boldsymbol{E}_{\text{inc}} \qquad \text{on } \Gamma_{\text{PEC}}. \tag{18}$$

Here $\boldsymbol{n}$ is the normal pointing out of the PEC object into the exterior medium. The DG mesh is a box with an inner spherical hole; the geometric boundary normal stored for the exterior computational domain points opposite to the PEC-object normal on the inner sphere. Therefore `build_pec_sphere_face_samples` flips the boundary-face normal before using it in the PEC residual and RCS current:

$$\boldsymbol{n}_{\text{RCS}} = -\boldsymbol{n}_{\text{DG,inner}}. \tag{19}$$

4.2 Absorbing Layer

The physical scattering problem is posed in an unbounded exterior domain. The production driver truncates the domain with a Cartesian nonlinear PML near the outer box boundaries. The RCS diagnostic is not evaluated on the outer PML; it is evaluated on the PEC sphere surface. A reliable RCS run therefore requires the PML to suppress outgoing scattered waves without reflecting them back to the sphere during the Fourier accumulation window.

5 Angular Coordinates

DiscoGMPI uses the standard spherical convention

$$\widehat{\boldsymbol{r}}(\theta, \phi) = \left(\sin\theta\cos\phi,\, \sin\theta\sin\phi,\, \cos\theta\right), \tag{20}$$

with

$$0 \le \theta \le \pi, \qquad 0 \le \phi < 2\pi.$$

Here θ is the polar scattering angle measured away from the positive z-axis, and ϕ is the azimuthal angle in the xy-plane measured from the positive x-axis toward the positive y-axis. The convention is shown in Figure 1. For the default incident wave propagating in $+z$, $\theta = 0°$ is forward scattering and $\theta = 180°$ is backscatter.

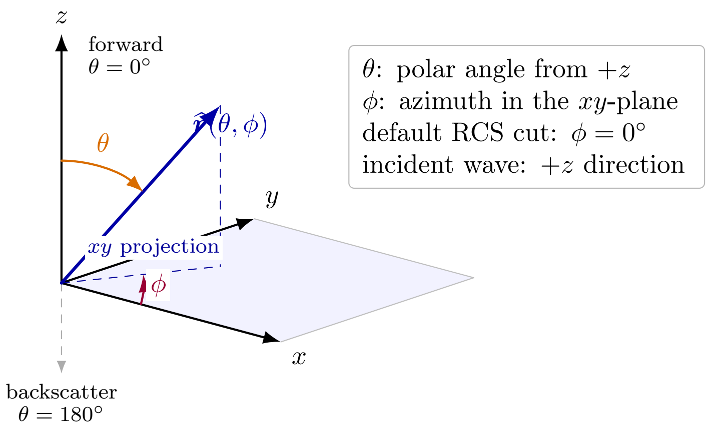

Figure 1: DiscoGMPI angular convention for RCS observation directions. The polar angle θ is measured from the $+z$ axis, while the azimuth ϕ is measured in the xy-plane from $+x$ toward $+y$.

The transverse basis vectors are

$$\widehat{\boldsymbol{e}}_\theta = (\cos\theta\cos\phi,\, \cos\theta\sin\phi,\, -\sin\theta)\,, \tag{21}$$
$$\widehat{\boldsymbol{e}}_\phi = (-\sin\phi,\, \cos\phi,\, 0)\,. \tag{22}$$

The reported far-field components are

$$E_\theta^\infty = \widehat{\boldsymbol{e}}_\theta \cdot \boldsymbol{E}^\infty, \qquad E_\phi^\infty = \widehat{\boldsymbol{e}}_\phi \cdot \boldsymbol{E}^\infty. \tag{23}$$

The RCS uses the transverse magnitude

$$|\boldsymbol{E}^\infty|^2 = |E_\theta^\infty|^2 + |E_\phi^\infty|^2. \tag{24}$$

For the default $+z$-propagating incident wave, $\theta = 0°$ is the forward direction and $\theta = 180°$ is the backscatter direction. The default RCS comparison uses the E-plane cut

$$\phi = 0°, \qquad 0° \le \theta \le 180°.$$

Some papers use a different naming convention for the angular variables. If a paper says "$\theta = 0$, $0 \le \phi \le 180°$" for a plot that varies from forward to backscatter, its ϕ is the physical scattering angle. In DiscoGMPI, the same physical plot is produced with

$$\phi = 0°, \qquad \theta \in [0°, 180°].$$

At $\theta = 0°$ and $\theta = 180°$, the azimuthal basis is geometrically degenerate: all values of ϕ describe the same Cartesian observation direction, but the transverse basis rotates. A $\theta = 0°$, $0 \le \phi \le 180°$ plot is therefore a basis orientation check, not the usual forward-to-backward RCS cut.

6 Near-To-Far Transformation

6.1 Outgoing Green Phase

With the phasor convention $\mathrm{e}^{-\mathrm{i}\omega t}$, the outgoing scalar Green function has the phase

$$G(\boldsymbol{x}, \boldsymbol{y}) = \frac{\mathrm{e}^{\mathrm{i}k|\boldsymbol{x}-\boldsymbol{y}|}}{4\pi|\boldsymbol{x}-\boldsymbol{y}|}. \tag{25}$$

For a far-field observation point $\boldsymbol{x} = r\widehat{\boldsymbol{r}}$,

$$|\boldsymbol{x} - \boldsymbol{y}| = r - \widehat{\boldsymbol{r}} \cdot \boldsymbol{y} + \mathcal{O}(r^{-1}). \tag{26}$$

Thus

$$\mathrm{e}^{\mathrm{i}k|\boldsymbol{x}-\boldsymbol{y}|} = \mathrm{e}^{\mathrm{i}kr}\mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{y}}\left(1 + \mathcal{O}(r^{-1})\right). \tag{27}$$

This is why the DiscoGMPI near-to-far surface integral uses

$$\exp(-\mathrm{i}k\,\widehat{\boldsymbol{r}} \cdot \boldsymbol{y}) \tag{28}$$

for the spatial phase factor. Using the opposite sign would evaluate the same Fourier transform in direction $-\widehat{\boldsymbol{r}}$, effectively reflecting an E-plane RCS curve by the mapping

$$\theta \mapsto 180^\circ - \theta.$$

6.2 PEC Equivalent Surface Current

For a PEC scatterer, the relevant equivalent electric surface current is

$$\widehat{\boldsymbol{J}}_s = \boldsymbol{n} \times \widehat{\boldsymbol{H}_{\mathrm{tot}}} \qquad \text{on } \Gamma_{\mathrm{PEC}}. \tag{29}$$

The code evaluates this using the total magnetic field

$$\boldsymbol{H}_{\mathrm{tot},h}(\boldsymbol{x}, t) = \boldsymbol{H}_{\mathrm{scat},h}(\boldsymbol{x}, t) + \boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x}, t). \tag{30}$$

The time-dependent surface-current integral for an observation direction $\widehat{\boldsymbol{r}}$ is

$$\boldsymbol{I}(\widehat{\boldsymbol{r}}, t) = \int_{\Gamma_{\mathrm{PEC}}} \boldsymbol{J}_s(\boldsymbol{y}, t)\mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{y}}\,\mathrm{d}S_y. \tag{31}$$

After Fourier extraction, the phasor integral is denoted $\widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})$.

6.3 Far-Field Reconstruction

The far-field amplitude used in DiscoGMPI is

$$\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) = \frac{\mathrm{i}kZ}{4\pi}\left[\widehat{\boldsymbol{r}}\left(\widehat{\boldsymbol{r}} \cdot \widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right) - \widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right]. \tag{32}$$

Equivalently,

$$\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) = \frac{\mathrm{i}kZ}{4\pi}\widehat{\boldsymbol{r}} \times \left(\widehat{\boldsymbol{r}} \times \widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right). \tag{33}$$

The vector in brackets is transverse because

$$\widehat{\boldsymbol{r}} \cdot \boldsymbol{E}^{\infty} = 0.$$

The overall sign convention depends on the normal convention and equivalent current convention. The RCS depends on $|\boldsymbol{E}^{\infty}|^2$, so a global sign does not affect the reported RCS.

Combining (2) and (32), the code reports

$$\sigma_h(\theta, \phi) = 4\pi\frac{|E^{\infty}_{\theta,h}|^2 + |E^{\infty}_{\phi,h}|^2}{A^2}. \tag{34}$$

7 Time-Domain Fourier Extraction

The solver advances real-valued fields in time. To obtain a monochromatic RCS, DiscoGMPI extracts the phasor at the incident frequency. If

$$f(t) = \mathrm{Re}\{\widehat{f}\, \mathrm{e}^{-\mathrm{i}\omega t}\},$$

then, over an integer number of periods,

$$\widehat{f} = \frac{2}{T} \int_0^T f(t)\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{d}t.$$

DiscoGMPI uses the discrete Hann-windowed estimator

$$\widehat{\boldsymbol{I}}_h(\widehat{\boldsymbol{r}}) = \frac{2}{W} \sum_m w_m\, \boldsymbol{I}_h(\widehat{\boldsymbol{r}}, t_m) \mathrm{e}^{\mathrm{i}\omega t_m}, \qquad W = \sum_m w_m. \tag{35}$$

The samples t_m are selected every `-rcs-every` time steps, and the final step is included. The current production drivers use a fixed adjusted time step, so the constant time-step factor cancels between numerator and denominator in (35).

The Hann window is

$$w(t) = \frac{1}{2} \left[1 - \cos\left(2\pi \frac{t - t_0}{T - t_0} \right) \right], \qquad t_0 \le t \le T, \tag{36}$$

where

$$t_0 = \texttt{-rcs-start-time}, \qquad T = \texttt{-final-time}.$$

Samples before t_0 are ignored. The purpose of t_0 is to exclude:

1. startup transients produced by abruptly imposing a time-harmonic incident field,
2. early nonperiodic adjustment of the scattered field,
3. wave packets reflected by the PML before they have decayed.

For a quantitative RCS comparison, the RCS curve should be stable when `-rcs-start-time` is shifted moderately later and when `-rcs-every` is reduced.

8 DG Surface Quadrature in the Code

The RCS calculation uses the same distributed DG mesh and polynomial basis as the time integration. The core data structure is `RCSFaceSample`, built by `build_pec_sphere_face_samples`. For each owned boundary face tagged as the metallic sphere:

1. The code keeps only faces with boundary tag `METALLIC_SPHERE_PEC_BOUNDARY_ID` = 10.
2. The face must belong to an element owned by the local MPI rank.
3. The face interpolation nodes are copied from the DG trace map.
4. The face quadrature weights are obtained by applying the reference face mass matrix to the constant vector 1, then scaling by physical face area:

$$w_q^{(F)} = \frac{|F|}{|F_{\mathrm{ref}}|} \left(M_F \mathbf{1} \right)_q. \tag{37}$$

5. The physical coordinates of each face node are reconstructed by mapping the reference tetrahedron coordinates to the physical tetrahedron.

6. The inner-boundary DG normal is flipped to obtain the PEC scatterer normal used in $\boldsymbol{J}_s = \boldsymbol{n} \times \boldsymbol{H}_{\mathrm{tot}}$.

At one time sample t_m, the local rank contribution is

$$\boldsymbol{I}_{h,\mathrm{rank}}(\widehat{\boldsymbol{r}}, t_m) = \sum_{F \in \mathcal{F}_{\mathrm{rank}}^{\mathrm{PEC}}} \sum_{q \in F} w_q^{(F)} \left[\boldsymbol{n}_F \times (\boldsymbol{H}_{\mathrm{scat},h}(\boldsymbol{x}_{Fq}, t_m) + \boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x}_{Fq}, t_m))\right] \mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{x}_{Fq}}. \tag{38}$$

The code accumulates this for every observation direction requested by the CLI. After multiplication by the Hann window and temporal Fourier phase, it adds the result to a rank-local complex accumulator:

$$\boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}) \leftarrow \boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}) + w_m \mathrm{e}^{\mathrm{i}\omega t_m} \boldsymbol{I}_{h,\mathrm{rank}}(\widehat{\boldsymbol{r}}, t_m). \tag{39}$$

The global accumulator is the MPI sum

$$\boldsymbol{A}(\widehat{\boldsymbol{r}}) = \sum_{\mathrm{rank}} \boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}). \tag{40}$$

Finally,

$$\widehat{\boldsymbol{I}}_h(\widehat{\boldsymbol{r}}) = \frac{2}{W} \boldsymbol{A}(\widehat{\boldsymbol{r}}). \tag{41}$$

9 Algorithm Implemented in DiscoGMPI

The RCS path in `distributed_metallic_sphere_scattering.jl` can be summarized as follows.

1. Parse the scattering configuration: mesh, sphere radius, wavelength, frequency, amplitude, DG order, ESPRK order, CFL, PML parameters, flux choice, RCS angles, RCS sampling interval, and output directory.

2. Build the incident wave parameters:

$$\boldsymbol{d} = \widehat{\boldsymbol{z}}, \qquad \boldsymbol{p} = \widehat{\boldsymbol{x}}, \qquad k = 2\pi/\lambda, \qquad \omega = 2\pi f.$$

3. Load and partition the tetrahedral box-minus-sphere mesh.

4. Build the DG discretization and the incident-aware PEC boundary data.

5. Initialize the numerical unknown as the scattered field.

6. If RCS is enabled, build:
   - observation directions $\widehat{\boldsymbol{r}}(\theta, \phi)$,
   - transverse bases $\widehat{\boldsymbol{e}}_\theta$, $\widehat{\boldsymbol{e}}_\phi$,
   - owned PEC face samples,
   - complex current accumulators.

7. March the scattered field in time with the selected Poisson-bracket Maxwell/ESPRK path and PML source terms.

8. At every RCS sample time:
   - set the boundary-data time consistently,
   - reconstruct $\boldsymbol{H}_{\mathrm{tot},h} = \boldsymbol{H}_{\mathrm{scat},h} + \boldsymbol{H}_{\mathrm{inc}}$ on the PEC face nodes,

- compute $\boldsymbol{J}_{s,h} = \boldsymbol{n} \times \boldsymbol{H}_{\mathrm{tot},h}$,
- integrate $\boldsymbol{J}_{s,h} \mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{x}}$ over the owned PEC faces,
- multiply by $w_m \mathrm{e}^{\mathrm{i}\omega t_m}$,
- add to the complex accumulator.

9. At final time, reduce the complex accumulators across MPI ranks.
10. Reconstruct $E^{\infty}_{\theta,h}$, $E^{\infty}_{\phi,h}$, and σ_h for every observation angle.
11. Write `diagnostics/rcs.csv`.

10 Observation Grid and CLI Controls

The production driver controls the RCS observation directions with:

`-rcs-theta-count`
: Number of polar angle samples between `-rcs-theta-min-degrees` and `-rcs-theta-max-degrees`.

`-rcs-theta-min-degrees`, `-rcs-theta-max-degrees`
: Minimum and maximum polar scattering angle in degrees.

`-rcs-phi-degrees`
: Comma-separated list of azimuthal angles in degrees.

`-rcs-every`
: Time-step interval between RCS surface integrations.

`-rcs-start-time`
: First physical time included in the Fourier extraction.

For a smooth E-plane RCS plot comparable to the common PEC-sphere benchmark, one should use, for example,

```
--rcs-theta-count=181 \
--rcs-theta-min-degrees=0 \
--rcs-theta-max-degrees=180 \
--rcs-phi-degrees=0
```

This gives one-degree angular spacing. Increasing `-rcs-theta-count` makes the RCS plot smoother but increases the RCS surface-integration cost approximately linearly. It does not change the DG time integration itself.

11 Output Files

The main RCS diagnostic file is

`<run-dir>/diagnostics/rcs.csv`.

It contains one row per observation direction. The key columns are:

Column	Meaning
theta_degrees, phi_degrees	Observation angles.
dir_x, dir_y, dir_z	Cartesian components of $\widehat{\boldsymbol{r}}$.
abs_E_infinity	$\lvert\boldsymbol{E}_h^\infty\rvert$.
abs_Etheta, abs_Ephi	$\lvert E_{\theta,h}^\infty\rvert$, $\lvert E_{\phi,h}^\infty\rvert$.
rcs	σ_h.
rcs_db	$10\log_{10}(\sigma_h)$, not normalized by πa^2.
real_Etheta, imag_Etheta	Complex $E_{\theta,h}^\infty$.
real_Ephi, imag_Ephi	Complex $E_{\phi,h}^\infty$.
samples	Number of nonzero-window RCS time samples.
window_sum	$W = \sum_m w_m$.
rcs_start_time, rcs_end_time	Fourier window endpoints.

The Python postprocessor writes:

- diagnostics/mie_rcs_comparison.csv,
- tables/mie_rcs_comparison.tex,
- plots/mie_rcs_comparison.pdf,
- plots/mie_rcs_polar.pdf,
- plots/mie_rcs_theta0_phi_cut.pdf.

The normalized plotted quantity is

$$10\log_{10}\left(\frac{\sigma}{\pi a^2}\right),$$

even though the raw rcs_db column in rcs.csv is the dB value of σ itself.

12 Example RCS Figures

Figures 2 and 3 show the postprocessed output from

output/validation_sequence/coarse_sphere_rcs_epw2_p4_transient_theta181/plots.

The file mie_rcs_theta0_phi_cut.pdf is intentionally not included here because, at $\theta = 0°$, varying ϕ rotates the transverse basis but does not change the Cartesian observation direction. The two plots included below are the physically relevant forward-to-backward E-plane RCS views for the default $\phi = 0°$ cut.

13 Exact PEC-Sphere Mie Reference

For a PEC sphere, the exact frequency-domain solution is given by the Mie series. The postprocessor evaluates the scattering amplitude functions $S_1(\theta)$ and $S_2(\theta)$ using the PEC Mie coefficients. With size parameter

$$x = ka,$$

the scalar Riccati-Bessel functions are

$$\psi_n(x) = xj_n(x), \qquad \xi_n(x) = xh_n^{(1)}(x),$$

and the PEC coefficients used by the code are

$$a_n = -\frac{\psi_n'(x)}{\xi_n'(x)}, \qquad b_n = -\frac{\psi_n(x)}{\xi_n(x)}. \tag{42}$$

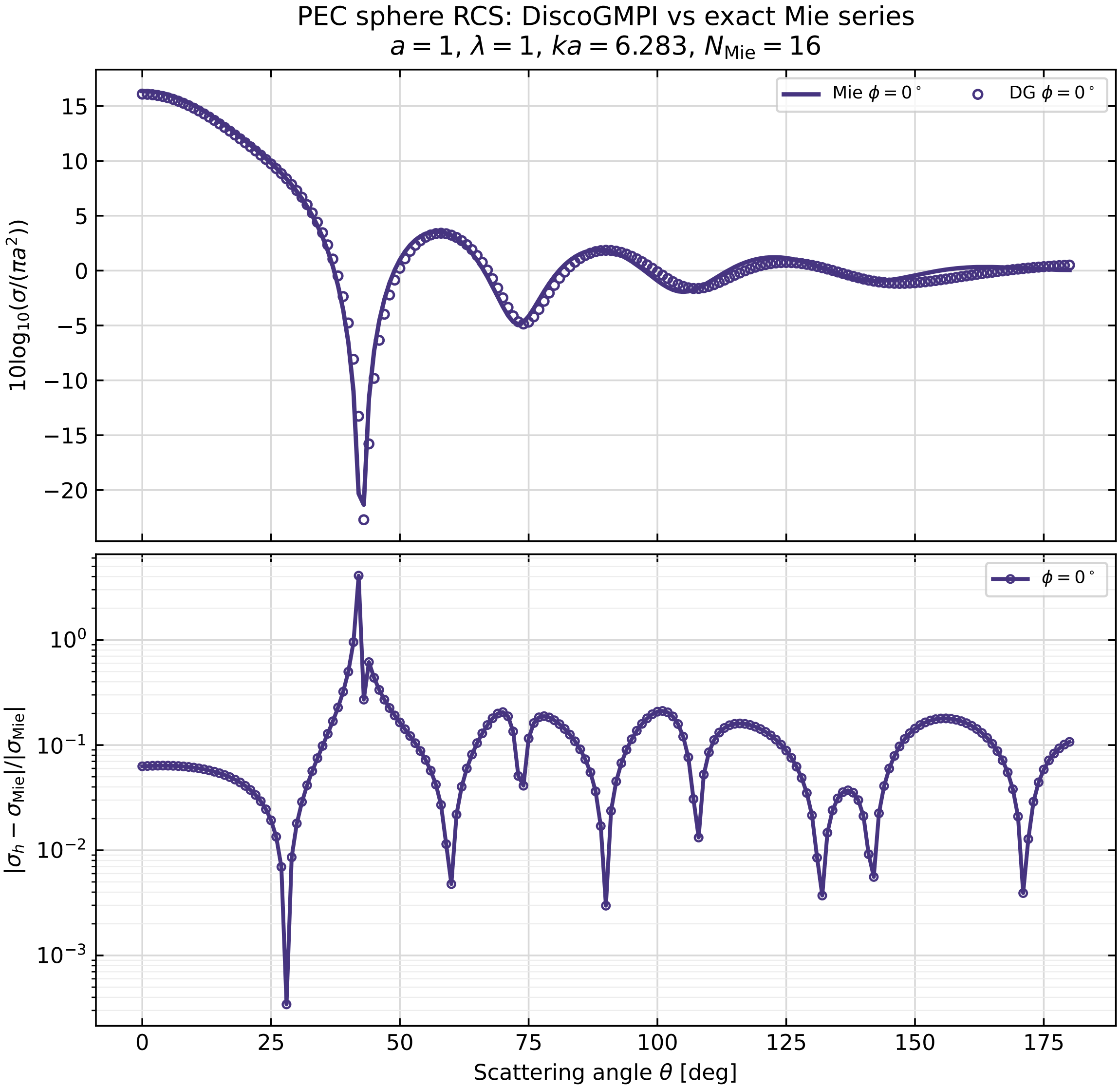

Figure 2: Cartesian RCS comparison for the `coarse_sphere_rcs_epw2_p4_transient_theta181` run. The upper panel compares $10\log_{10}(\sigma/(\pi a^2))$ for the DG near-to-far extraction and the exact PEC-sphere Mie series. The lower panel reports the relative RCS error.

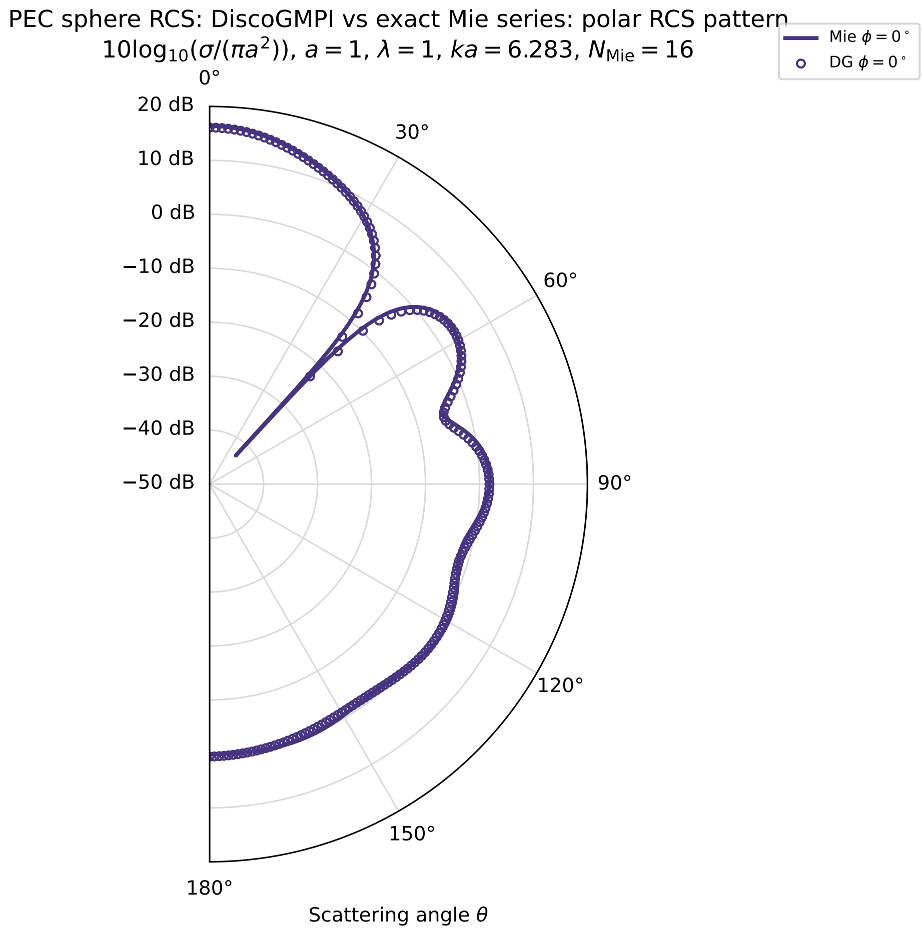

Figure 3: Polar RCS pattern for the same run. The angular coordinate is the DiscoGMPI polar scattering angle θ, with $\theta = 0^\circ$ denoting the forward $+z$ direction and $\theta = 180^\circ$ denoting backscatter. The radial tick labels are normalized dB values, $10\log_{10}(\sigma/(\pi a^2))$.

The angular functions are

$$\pi_1(\mu) = 1, \tag{43}$$
$$\pi_2(\mu) = 3\mu, \tag{44}$$
$$\pi_n(\mu) = \frac{2n-1}{n-1}\mu\,\pi_{n-1}(\mu) - \frac{n}{n-1}\pi_{n-2}(\mu), \qquad n \geq 3, \tag{45}$$
$$\tau_n(\mu) = n\mu\,\pi_n(\mu) - (n+1)\pi_{n-1}(\mu), \qquad \mu = \cos\theta. \tag{46}$$

The scattering amplitudes are

$$S_1(\theta) = \sum_{n=1}^{N} \frac{2n+1}{n(n+1)} \left[a_n \pi_n(\cos\theta) + b_n \tau_n(\cos\theta)\right], \tag{47}$$
$$S_2(\theta) = \sum_{n=1}^{N} \frac{2n+1}{n(n+1)} \left[a_n \tau_n(\cos\theta) + b_n \pi_n(\cos\theta)\right]. \tag{48}$$

For the DiscoGMPI default incident wave, the postprocessor maps these amplitudes to the same $(\widehat{\boldsymbol{e}}_\theta, \widehat{\boldsymbol{e}}_\phi)$ basis as the numerical RCS:

$$E^{\infty}_{\theta,\mathrm{Mie}} = \frac{S_2(\theta)\cos\phi}{k}, \qquad E^{\infty}_{\phi,\mathrm{Mie}} = -\frac{S_1(\theta)\sin\phi}{k}. \tag{49}$$

Then

$$\sigma_{\mathrm{Mie}}(\theta,\phi) = 4\pi \left(|E^{\infty}_{\theta,\mathrm{Mie}}|^2 + |E^{\infty}_{\phi,\mathrm{Mie}}|^2\right). \tag{50}$$

The automatic truncation order is

$$N = \max\left(8,\ \left\lceil x + 4x^{1/3} + 2 \right\rceil\right). \tag{51}$$

14 Validation Strategy

A correct RCS workflow must pass several levels of validation.

14.1 Boundary Residual

Before trusting RCS data, verify the incident-aware PEC boundary condition:

$$R_{\mathrm{PEC}} = \left\| \boldsymbol{n} \times (\boldsymbol{E}_{\mathrm{scat},h} + \boldsymbol{E}_{\mathrm{inc}}) \right\|_{L^2(\Gamma_{\mathrm{PEC}})}. \tag{52}$$

The validation driver `examples/validate_pec_sphere_boundary_residual.jl` writes `diagnostics/pec_boun` A large residual means the surface current is being evaluated on a field that does not satisfy the PEC condition accurately.

14.2 Qualitative Near Field

Inspect $\boldsymbol{E}_{\mathrm{scat}}$, $\boldsymbol{H}_{\mathrm{scat}}$, $\boldsymbol{E}_{\mathrm{tot}}$, and $\boldsymbol{H}_{\mathrm{tot}}$ in ParaView. The production driver can write:

- $\boldsymbol{E}_{\mathrm{scat}}, \boldsymbol{H}_{\mathrm{scat}}$,
- $\boldsymbol{E}_{\mathrm{inc}}, \boldsymbol{H}_{\mathrm{inc}}$,
- $\boldsymbol{E}_{\mathrm{tot}}, \boldsymbol{H}_{\mathrm{tot}}$,
- $|\boldsymbol{E}_{\mathrm{scat}}|, |\boldsymbol{H}_{\mathrm{scat}}|$.

For the default plane wave, the components E_x and H_y are the most direct comparison with the metallic-sphere benchmark figures.

14.3 Mie Comparison

After a run finishes, use

```
python3 examples/postprocess_metallic_sphere_mie_rcs.py \
  --run-dir <run-dir>
```

The comparison figure should be interpreted as:

- top panel: numerical and Mie $10\log_{10}(\sigma/(\pi a^2))$;
- bottom panel:

$$\frac{|\sigma_h - \sigma_{\mathrm{Mie}}|}{|\sigma_{\mathrm{Mie}}|}.$$

For quantitative validation, repeat the run with:

1. later `-rcs-start-time`,
2. smaller `-rcs-every`,
3. more angular samples,
4. finer sphere geometry,
5. larger polynomial order,
6. lower PML reflection.

A reliable RCS estimate should not change significantly under these variations.

15 Recommended Commands

A high-order EPW2/EPW4 smoke or development run can be launched with:

```
mpiexec -n 2 -genv OPENBLAS_NUM_THREADS 1 julia --project=. \
  examples/validate_coarse_sphere_rcs.jl \
  --mesh=examples/meshes/metallic_sphere_scattering_epw2.vtk \
  --order=4 --esprk-order=5 \
  --final-time=12.0 --rcs-start-time=5.0 \
  --rcs-every=5 --diagnostics-every=250 \
  --rcs-theta-count=181 \
  --rcs-theta-min-degrees=0 \
  --rcs-theta-max-degrees=180 \
  --rcs-phi-degrees=0 \
  --min-max-rcs=0 \
  --output-dir=output/validation_sequence/sphere_rcs_epw2_p4_theta181
```

Then postprocess:

```
python3 examples/postprocess_metallic_sphere_mie_rcs.py \
  --run-dir output/validation_sequence/sphere_rcs_epw2_p4_theta181
```

For a publication-quality RCS curve, EPW2 is usually too coarse geometrically. Use at least EPW3 or EPW4, enough runtime for a steady monochromatic response, and an RCS window that excludes startup transients.

16 Common Failure Modes

Too few points in the RCS curve.
Increase `-rcs-theta-count`. The postprocessor only plots the angles present in `diagnostics/rcs.csv`.

Mie and DG curves are mirrored left-to-right.
Check the near-to-far phase sign. For the convention $\cos(kz - \omega t)$, the surface integral uses $\exp(-\mathrm{i}k\widehat{\boldsymbol{r}} \cdot \boldsymbol{x})$.

Large error near deep Mie nulls.
Relative error is sensitive when σ_{Mie} is very small. Inspect absolute normalized error and dB error as well.

Good boundary residual but poor RCS.
Possible causes are insufficient sphere geometry resolution, PML reflection, poor Fourier window, too few periods in the accumulation window, or too coarse angular sampling.

Good early RCS but later contamination.
Reflections from the PML or outer boundary may be returning to the sphere. Increase PML quality, enlarge the buffer, or end the Fourier window before reflections return.

No RCS rows.
RCS was disabled, the run failed before final output, no faces were tagged with the metallic sphere boundary ID, or the Fourier window contained no nonzero samples.

Confusion between θ and ϕ.
In DiscoGMPI, θ is the polar scattering angle from $+z$, and ϕ is the azimuthal angle. The standard E-plane sphere plot is $\phi = 0$ with $\theta \in [0°, 180°]$.

17 Checklist for a Quantitative RCS Study

1. Verify the mesh: sphere boundary tag 10, adequate sphere surface resolution, adequate PML/buffer region.
2. Verify the incident wave: direction $+z$, polarization x, wavelength, frequency, amplitude, ϵ, and μ.
3. Verify PEC residual: $\|\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{tot}}\|_{L^2(\Gamma_{\mathrm{PEC}})}$.
4. Run long enough for a periodic scattered field.
5. Set `-rcs-start-time` after the startup transient.
6. Use enough angular samples, e.g. `-rcs-theta-count=181` or `361`.
7. Compare against the corrected Mie postprocessor.
8. Repeat with a later `-rcs-start-time`.
9. Repeat with a smaller `-rcs-every`.
10. Repeat on a finer mesh or with a higher-order discretization.
11. Confirm that the RCS curve and error metrics stabilize.

18 Code Map

The main implementation locations are:

File or function	Role
`examples/distributed_metallic_sphere_scattering.jl`	Production scattering and RCS driver.
`RCSObservationDirection`	Stores (θ, ϕ), $\widehat{\boldsymbol{r}}$, $\widehat{\boldsymbol{e}}_\theta$, $\widehat{\boldsymbol{e}}_\phi$.
`RCSFaceSample`	Stores owned PEC face nodes, weights, coordinates, and normals.
`build_pec_sphere_face_samples`	Extracts tagged sphere faces and flips the inner-boundary normal.
`accumulate_rcs_sample!`	Computes the windowed surface-current Fourier contribution.
`global_rcs_accumulator`	MPI reduction of complex accumulators.
`far_field_from_surface_current`	Applies (32).
`write_rcs_results`	Writes `diagnostics/rcs.csv`.
`src/MaxwellExactSolutions.jl`	Incident plane-wave definitions.
`examples/validate_coarse_sphere_rcs.jl`	Coarse validation gate and Julia Mie comparison.
`examples/postprocess_metallic_sphere_mie_rcs.py`	Publication-style Mie comparison plots and tables.

19 Summary

DiscoGMPI computes PEC-sphere RCS by solving the scattered-field Maxwell problem in the time domain, reconstructing the total magnetic field on the metallic surface, integrating the equivalent PEC surface current with the outgoing far-field phase $\exp(-\mathrm{i}k\widehat{\boldsymbol{r}} \cdot \boldsymbol{x})$, extracting the phasor at the incident frequency with a Hann-windowed Fourier average, and normalizing the far-field amplitude according to

$$\sigma = 4\pi|\boldsymbol{E}^\infty|^2/A^2.$$

The procedure is sensitive to phase convention, angular convention, surface normal orientation, mesh geometry, PML reflections, and Fourier-window selection. The corrected Mie postprocessor provides the primary quantitative reference for the PEC sphere, but a trustworthy comparison requires a resolved mesh and a transient-free Fourier window.